

Inside a Cyber Attack

How Hackers Think & How SOC Defends

- Cybersecurity Awareness Session
- Security Operations Perspective

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Agenda

- Introduction to Cyber Threat Landscape
- Understanding Hacker Mindset
- Cyber Attack Journey (End-to-End View)
- Role of SOC in Detection & Defense
- Key Challenges in Cybersecurity
- Summary & Key Takeaways

Cyber Threat Landscape

Current trends and emerging cyber threats



- 01 **Increasing Attacks**
Significant rise in ransomware and targeted attacks
- 02 **Evolving Methods**
Increased use of automation and AI-driven techniques
- 03 **Expanding Surface**
Expansion of attack surface across cloud, endpoints, identities





Why This Matters

Business Impact of Cyber Attacks



- Cyber attacks impact business continuity and operations
- Financial and reputational risks are increasing
- Delayed detection leads to higher damage and recovery cost
- Proactive monitoring is essential for resilience

Attacker *Objectives*

Primary Objectives of Threat Actors



- Financial gain through ransomware or fraud
- Unauthorized access to sensitive data
- Long-term persistence within systems
- Disruption of critical business services





Hacker

Mindset

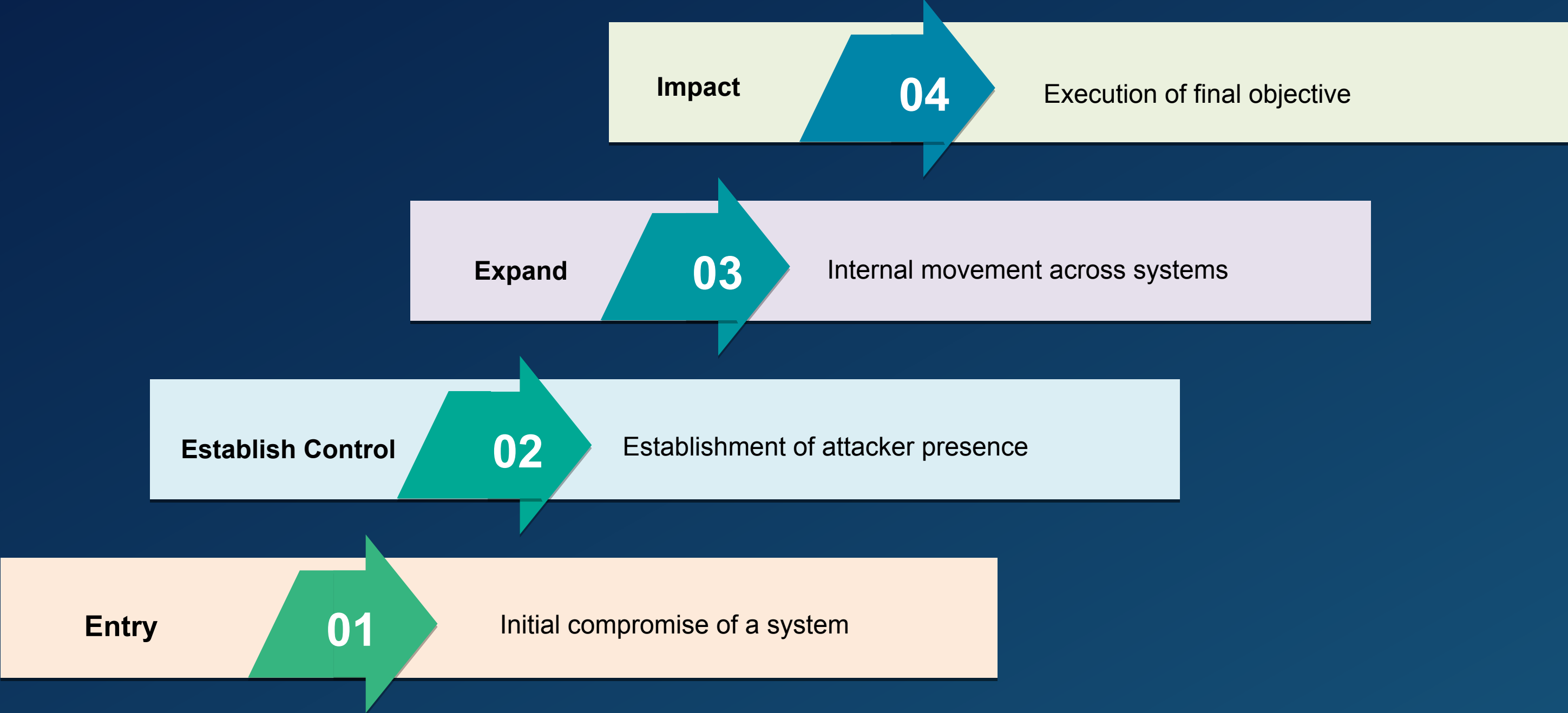
How Attackers Think and Operate



- Identify and exploit the weakest entry point
- Maintain stealth to avoid detection
- Gradually escalate privileges
- Move laterally to expand access

Cyber Attack Journey

Stages of a Cyber Attack Lifecycle ↗





Common Entry Points

How Attackers Gain Initial Access ↗

- Phishing and social engineering attacks
- Weak or compromised credentials
- Exploited software vulnerabilities
- Misconfigured systems and services

Post -Compromise *Activities*

What Happens After Initial Access ↗

- Malware execution and persistence setup
- Establishment of command & control communication
- Credential harvesting and system discovery
- Continuous monitoring of environment





Lateral Movement

Lateral Movement Techniques Used by Attackers



- Movement across systems within the network
- Use of legitimate tools to avoid detection
- Expansion of access using stolen credentials
- Targeting high-value systems and assets

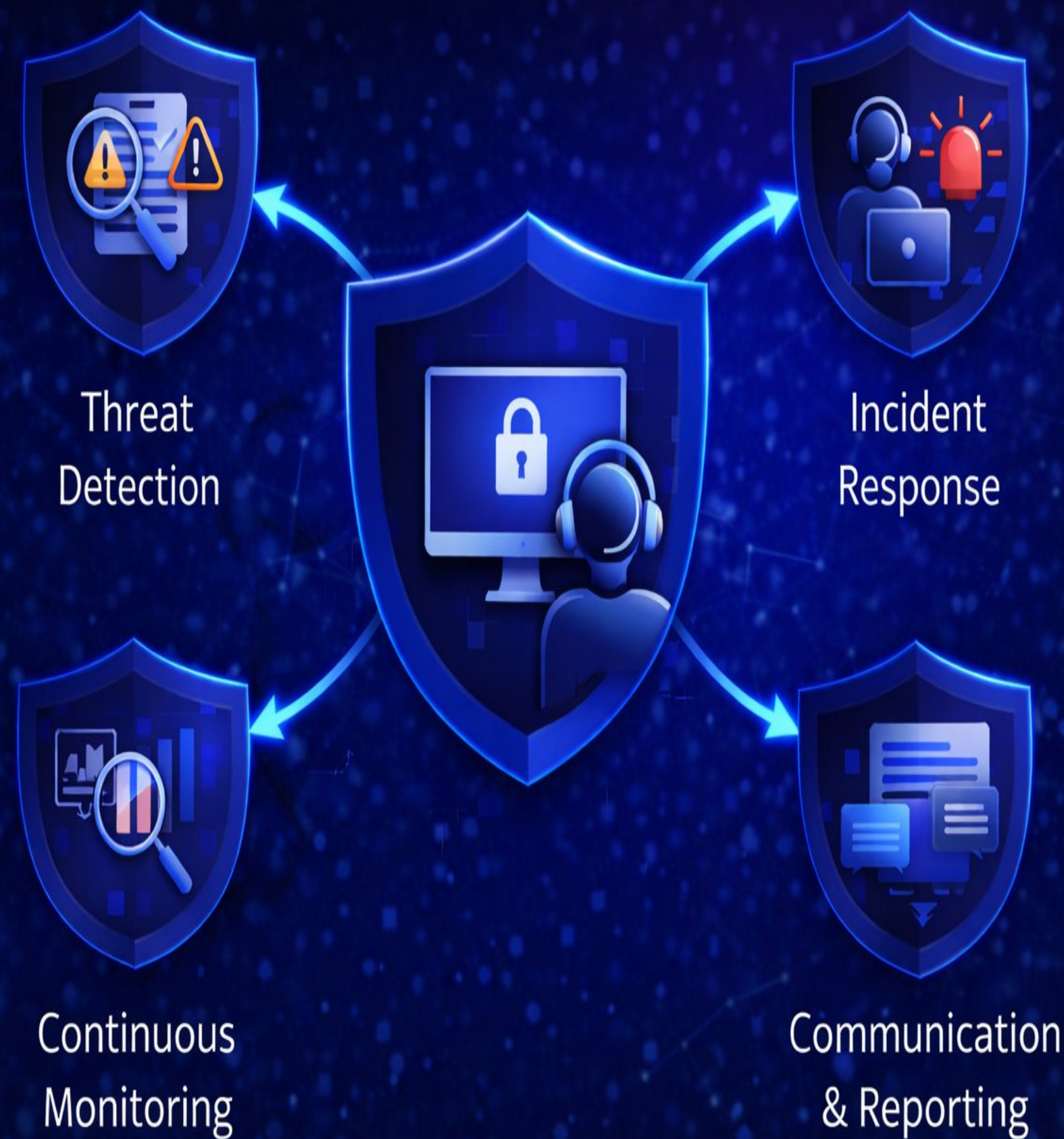
*Final*Impact

Impact of a Successful Cyber Attack



- Data exfiltration of sensitive information
- Deployment of ransomware or destructive malware
- Disruption of business operations
- Regulatory and reputational consequences





Role of **SOC**

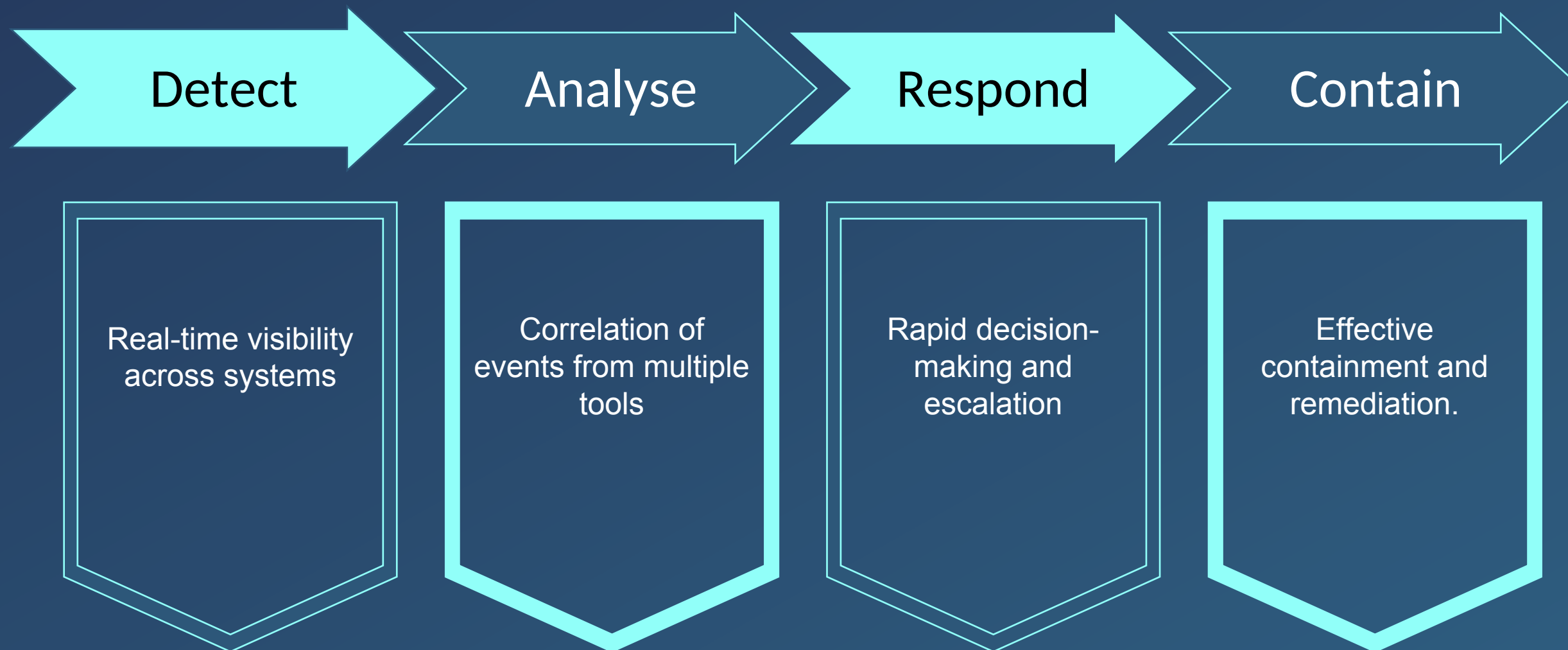
Core Functions of a Security Operations Center ↗

- Continuous 24/7 monitoring of security events
- Detection of suspicious activities and anomalies
- Investigation and validation of alerts
- Coordinated response to security incidents

SOC – Défense

Approach

SOC Incident Response Workflow ↗



Speed of Detection

Why Speed Matters in Threat Detection



- Faster detection reduces overall impact
- Delayed response increases attack spread
- Automation improves response efficiency
- Visibility enables proactive defense





Key Challenge

Key Challenges in Modern Cybersecurity Operations



- High volume of alerts and false positives
- Increasing sophistication of threats
- Limited visibility across distributed environments
- Dependency on multiple tools and integrations

Summary & Takeaways

Key Takeaways from Cyber Attack Analysis



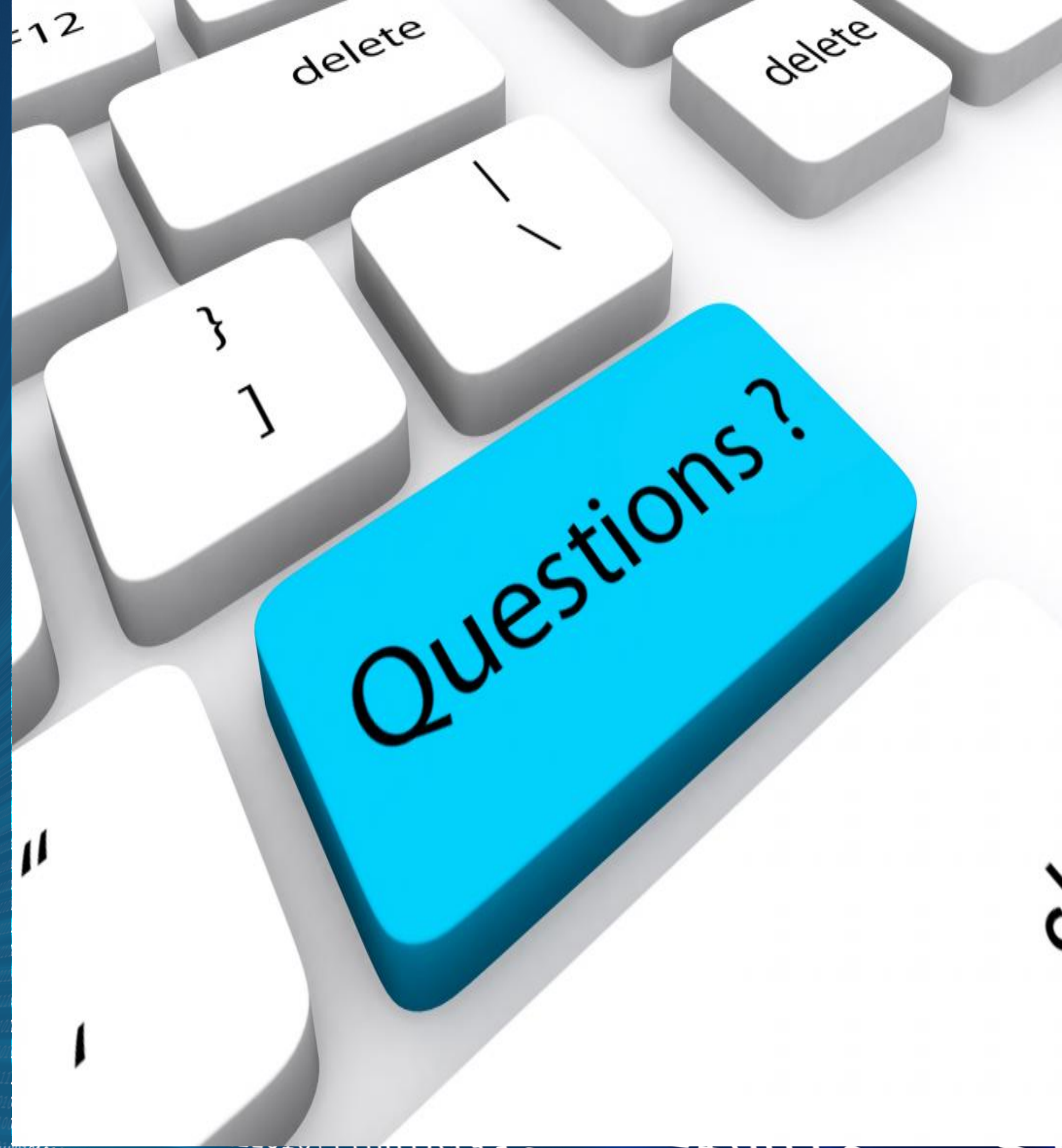
- Cyber attacks follow structured patterns
- Early detection significantly reduces impact
- SOC is critical for real-time defense
- Continuous improvement strengthens security posture

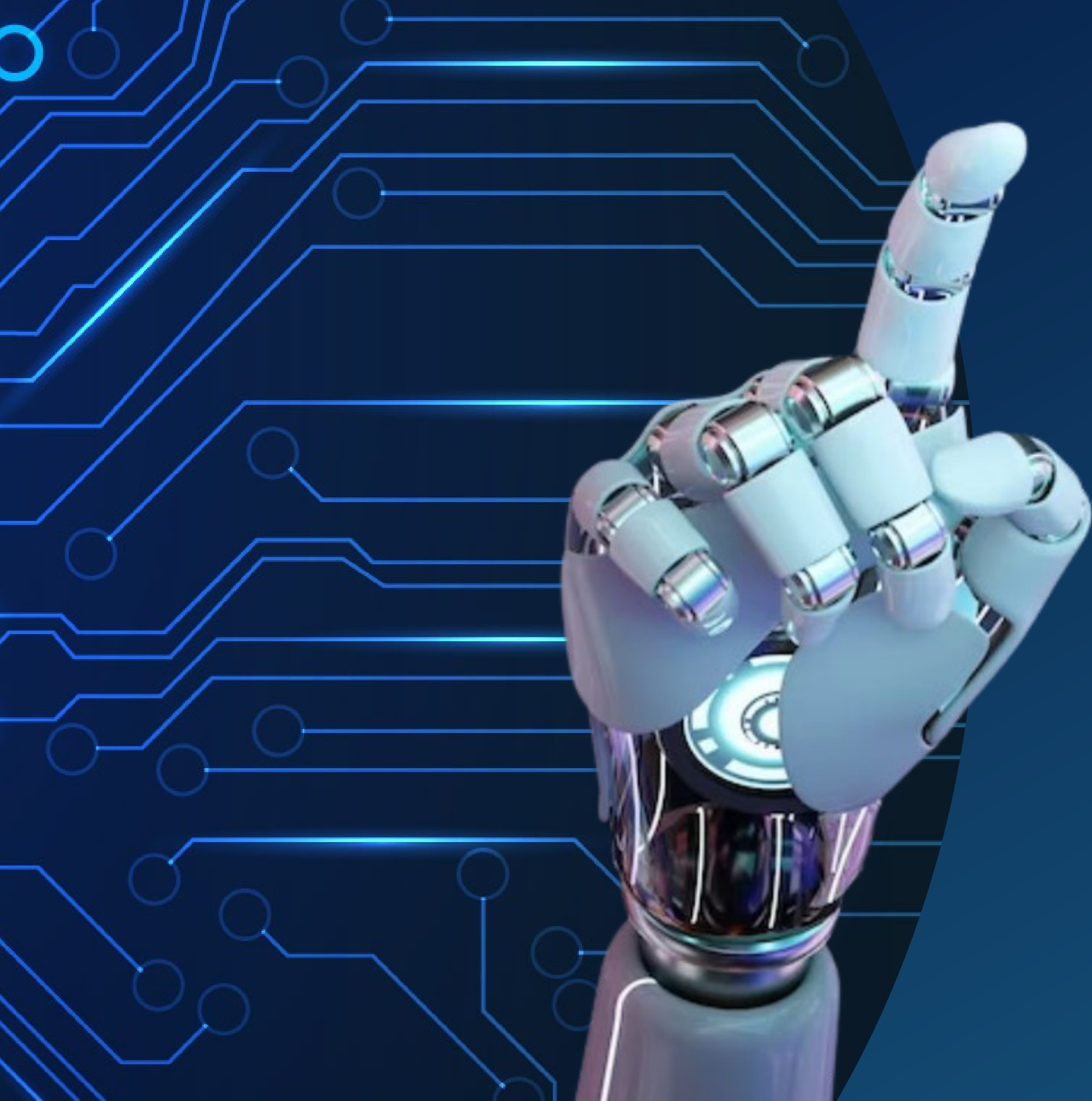


Any

Open Floor for Queries

Clarifications & Insights





***Thank
You!***

Appreciate your time and attention

Stay Secure | Stay Vigilant

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